What are IP, MAC, DNS, DHCP and ports? Include a table of 12 common ports: 21, 22, 23, 25, 53, 80, 110, 143, 443, 445, 3306, 3389 with service and typical risk.

Networking Foundations & Ports

- **IP Address**: A logical software address used to route data to a device on a network (e.g., 192.168.1.50).
- **MAC Address**: A permanent, unique physical hardware address burned into a device's network card at the factory (e.g., 00:1A:2B:3C:4D:5E).
- **DNS**: Translates human-readable domain names (google.com) into machine-readable IP addresses.
- **DHCP**: Automatically assigns IP addresses and network settings to devices when they connect.
- **Port**: A logical endpoint (numbered 0 to 65535) used to direct network traffic to a specific service running on a machine.

12 Common Ports and Risks

Port	Service	Typical Risk / Exploit
21	FTP	Transmits data and credentials in cleartext; vulnerable to sniffing.
22	SSH	Targeted heavily for password brute-forcing or software flaws.
23	Telnet	Completely unencrypted text; highly vulnerable to credential theft.
25	SMTP	Can be abused for email spamming, phishing, or user enumeration.
53	DNS	Vulnerable to cache poisoning and DDoS amplification attacks.
80	HTTP	Unencrypted web traffic; credentials and data can be sniffed.
110	POP3	Cleartext email retrieval; exposes passwords.
143	IMAP	Cleartext email synchronization; exposes passwords.

Port	Service	Typical Risk / Exploit
443	HTTPS	Encrypted web traffic, but web flaws (SQLi, XSS) still work inside the tunnel.
445	SMB	Used for file sharing; famous target for remote code exploits (like EternalBlue).
3306	MySQL	Direct database access; vulnerable to brute-force and remote data theft.
3389	RDP	Remote desktop access; heavily targeted for brute-force attacks and system takeovers.